

NittanyAuction Relational Schema

Please find the schemas which you will use for the **NittanyAuction** prototype below. You should use your judgement for assigning proper datatypes and integrity constraints (recommended primary keys in red). You may also augment the schemas if you feel that helps with the implementation of new features. In that case, you will also need to prepare for the corresponding data for testing and final demonstration.

Users(email, password)

- All the users in the system and their passwords.

Helpdesk(email, position)

- All the helpdesk staffs and their positions.

Requests(request_id, sender_email, helpdesk_staff_email, request_type, request_desc, request_status)

- Users send requests to the team of helpdesk staffs, initially assigned to a psuedo staff helpdeskteam@lsu.edu.
- A helpdesk_staff_email different from helpdeskteam@lsu.edu indicates the staff working on the request.
- Request_type indicates the types of requests, e.g., ChangeID, AddCategory, MarketAnalysis, etc.
- Request_desc describes the requests.
- Request_status indicates whether a request is completed or not. (0: incomplete; 1: completed)

Bidders(email, first_name, last_name, age, home_address_id, major)

- Bidders is a subset of the Users.

Credit_Cards(credit_card_num, card type, expire_month, expire_year, security_code, Owner_email)

- Credit cards stored in the system.
- A credit card is owned by one bidder only.
- An owner (bidder) may have one or multiple credit cards stored in the system.

Address(address_ID, zipcode, street_num, street_name)

- The home_address_id in Bidders table and the business_address_id in Local_Vendors table store address_ID existing in this Address table (i.e., foreign key).

Zipcode_Info(zipcode, city, state)

- This table stores the city and state of a zipcode area.

Sellers(email, bank_routing_number, bank_account_number, balance)

- Sellers consist of a subset of bidders and all local vendors.

Local_Vendors(Email, Business_Name, Business_Address_ID, Customer_Service_Phone_Number)

- Local_vendors is a subclass of Sellers, i.e., all local vendors are sellers.

Categories(parent_category, category_name)

- This table stores the links between a category and its parent category in the category hierarchy.
- A category has only one parent category, but its parent category may have multiple children.

Auction_Listings(Seller Email, Listing_ID, Category, Auction_Title, Product_Name, Product_Description, Quantity, Reserve_Price, Max_bids, Status)

- An Auction_Listing (also referred to as Product_Listing in the project description) records useful information about the auction of a listed product).
- Each seller has their own counter of Listing_ID, i.e., Listing_ID needs to be unique under the same seller. Therefore, Seller Email and Listing_ID in combination serve as the primary key.
- Auction_Title, Product_Name and Product Description are information about the product listed for auction. You may use them as appropriate in your display of auction information.
- A reserve price is a minimum price that a seller would be willing to accept from a buyer. If the reserve price is not met, the seller is not required to sell the item, even to the highest bidder.
- For ease of demonstration, we use the number of biddings (instead of a stop bidding time) to specify the end of an auction. Max_bids is the maximum number of bids for an auction. The auction ends when Max_bids is reached.
- An auction_listing has the status “active” (i.e., 1) by default. When a seller wants to take off an auction_listing from the market, the status of the auction_listing is changed to “inactive” (i.e., 0). After a product_listing is sold, the status is “sold” (i.e., 2).

Bids(Bid_ID, Seller_Email, Listing_ID, Bidder_email, Bid_price)

- Each bid has a unique Bid_ID generated by the system during bidding.
- The price of a new bid needs to be at least \$1 higher than all previous bids

Transactions(Transaction_ID, Seller_Email, Listing_ID, Buyer_Email, Date, Payment)

- The Transactions table records the payment after a successful bidding.
- Transaction_ID uniquely identifies the transaction.
- The auction listings in the Transactions table, identified by (Seller_Email, Listing_ID), are a subset of those in the Auction_Listings table.

Rating(Bidder_Email, Seller_Email, Date, Rating, Rating_Desc)

- The Rating table contains historic rating information collected from other sources. It covers more than what's maintained in the transaction table.
- A bidder needs to bid successfully for (and buy) a product listing from a seller in order to rate the seller.
- A bidder can only rate the seller once on a given date.